

PHYS 3316 — Quantum Mechanics I

Notebook 2 — The Quantum Harmonic Oscillator

Instructor: Eric R. Bittner

Department of Physics, University of Houston

Overview

The quantum harmonic oscillator is one of the most important exactly solvable models in quantum mechanics. It provides a useful approximation for systems undergoing small oscillations near a stable equilibrium and appears throughout atomic, molecular, optical, and condensed-matter physics.

The Hamiltonian is

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2.$$

Rather than solving the Schrödinger equation directly as a differential equation, we will develop the problem primarily using the **raising and lowering operator method**.

The harmonic oscillator also provides a natural introduction to the **variational theorem**. A Gaussian trial wavefunction can be optimized to minimize the expectation value of the energy. For the quadratic potential, this procedure reproduces the exact ground state.

Learning Objectives

After completing this notebook, you should be able to:

1. Define the harmonic-oscillator raising and lowering operators.
2. Use operator algebra to determine the energy spectrum.
3. Calculate matrix elements of position and momentum.
4. Construct and plot harmonic-oscillator eigenfunctions.
5. Verify normalization and orthogonality numerically.
6. Compute expectation values and uncertainties.
7. Construct and evolve superpositions of oscillator eigenstates.
8. Compare classical and quantum phase-space motion.
9. Apply the variational theorem using a Gaussian trial function.

10. Interpret variational minimization as a competition between kinetic and potential energy.

Physical Model

The harmonic-oscillator potential is

$$V(x) = \frac{1}{2}m\omega^2x^2,$$

where

- m is the particle mass,
- ω is the angular frequency, and
- $x = 0$ is the equilibrium position.

The potential is symmetric about the origin and increases quadratically as the particle moves away from equilibrium.

Classically, the particle oscillates between the turning points

$$x_{\text{tp}} = \pm \sqrt{\frac{2E}{m\omega^2}}.$$

Quantum mechanically, the allowed energies are discrete:

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots$$

The lowest possible energy is therefore

$$E_0 = \frac{1}{2}\hbar\omega.$$

This nonzero ground-state energy is called the **zero-point energy**.

Natural Harmonic-Oscillator Units

The harmonic oscillator defines natural scales for position, momentum, energy, and time.

The characteristic length is

$$x_0 = \sqrt{\frac{\hbar}{m\omega}}.$$

The characteristic momentum is

$$p_0 = \sqrt{m\hbar\omega}.$$

The natural energy scale is

$$E_0 = \hbar\omega,$$

and the natural time scale is

$$t_0 = \frac{1}{\omega}.$$

It is often convenient to introduce the dimensionless coordinate

$$\xi = \frac{x}{x_0} = \sqrt{\frac{m\omega}{\hbar}} x.$$

In terms of ξ , the Hamiltonian becomes

$$\frac{\hat{H}}{\hbar\omega} = -\frac{1}{2} \frac{d^2}{d\xi^2} + \frac{1}{2} \xi^2.$$

Unless otherwise stated, the computational sections of this notebook use dimensionless harmonic-oscillator units:

$$\hbar = 1, \quad m = 1, \quad \omega = 1.$$

With this convention,

$$\hat{H} = -\frac{1}{2} \frac{d^2}{dx^2} + \frac{1}{2} x^2$$

and

$$E_n = n + \frac{1}{2}.$$

Raising and Lowering Operators

Define the lowering operator

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \hat{x} + \frac{i}{\sqrt{2m\hbar\omega}} \hat{p},$$

and the raising operator

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \hat{x} - \frac{i}{\sqrt{2m\hbar\omega}} \hat{p}.$$

Note that the text book uses a rather unstandard notation of a_- for the lowering operator and a_+ as the raising operator. In my lectures and notes,

I'll use the more commonly adapted notation used here.

These operators satisfy the commutation relation

$$[\hat{a}, \hat{a}^\dagger] = 1.$$

The Hamiltonian can then be written as

$$\hat{H} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right).$$

The operators act on the energy eigenstates according to

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle,$$

and

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle.$$

The position and momentum operators can be expressed as

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger),$$

and

$$\hat{p} = -i\sqrt{\frac{m\hbar\omega}{2}} (\hat{a} - \hat{a}^\dagger).$$

These relations make it possible to calculate many expectation values and matrix elements without directly evaluating integrals.

Stationary-State Wavefunctions

The normalized position-space eigenfunctions are

$$\psi_n(x) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} H_n(\xi) e^{-\xi^2/2},$$

where

$$\xi = \sqrt{\frac{m\omega}{\hbar}} x$$

and $H_n(\xi)$ is the Hermite polynomial of degree n .

The first few Hermite polynomials are

$$H_0(\xi) = 1,$$

$$H_1(\xi) = 2\xi,$$

$$H_2(\xi) = 4\xi^2 - 2,$$

$$H_3(\xi) = 8\xi^3 - 12\xi.$$

The parity of the eigenfunctions alternates:

$$\psi_n(-x) = (-1)^n \psi_n(x).$$

Therefore,

- even n states have even parity,
- odd n states have odd parity.

The state ψ_n has exactly n nodes.

Variational Principle

The variational theorem states that for any normalized trial wavefunction ϕ ,

$$E[\phi] = \langle \phi | \hat{H} | \phi \rangle \geq E_0,$$

where E_0 is the exact ground-state energy.

A trial wavefunction may contain one or more adjustable parameters. The best approximation within the chosen family of trial functions is obtained by minimizing the energy with respect to those parameters.

For the harmonic oscillator, we will use a normalized Gaussian trial function,

$$\phi_\alpha(x) = \left(\frac{2\alpha}{\pi} \right)^{1/4} e^{-\alpha x^2}, \quad \alpha > 0.$$

The variational energy is

$$E(\alpha) = \langle \phi_\alpha | \hat{H} | \phi_\alpha \rangle.$$

The parameter α controls the width of the Gaussian:

- large α produces a narrow wavefunction,
- small α produces a broad wavefunction.

The optimum width results from a balance between kinetic and potential energy.

Computational Workflow

For each computational exercise, follow the sequence

derive $\rightarrow$ implement $\rightarrow$ verify $\rightarrow$ interpret $\rightarrow$ explore.

Each problem may contain the following components:

Physical Objective

Identify the physical question being investigated.

Mathematical Background

Write the relevant equations before implementing them.

Computational Task

Translate the mathematical expressions into Python.

Expected Output

Generate plots, tables, or numerical comparisons.

Physical Interpretation

Explain what the results mean physically.

Challenge

Modify the calculation or extend it to a more general system.

Python Packages

The notebook will use the following packages:

```
import numpy as np
import matplotlib.pyplot as plt

from scipy.integrate import simpson
from scipy.optimize import minimize_scalar
from scipy.special import eval_hermite, factorial
```

Problem 1 — Raising and Lowering Operators

The lowering and raising operators for the one-dimensional harmonic oscillator are

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \hat{x} + \frac{i}{\sqrt{2m\hbar\omega}} \hat{p},$$

and

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \hat{x} - \frac{i}{\sqrt{2m\hbar\omega}} \hat{p}.$$

The Hamiltonian is

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2 \hat{x}^2.$$

(a)

Show that

$$[\hat{a}, \hat{a}^\dagger] = 1.$$

(b)

Express the Hamiltonian in terms of the ladder operators and show that

$$\hat{H} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right).$$

(c)

Suppose

$$\hat{a}|0\rangle = 0.$$

Determine the ground-state energy.

(d)

Show that repeated application of the raising operator generates an infinite sequence of normalized eigenstates,

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}} |0\rangle.$$

Hence determine the complete energy spectrum.

(e)

Explain why the harmonic oscillator possesses a nonzero ground-state energy. Relate your answer to the uncertainty principle.

SOLUTION:

Problem 2 — Matrix Elements and Selection Rules

Using the ladder operators, show that

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger)$$

and

$$\hat{p} = -i\sqrt{\frac{m\hbar\omega}{2}}(\hat{a} - \hat{a}^\dagger).$$

(a)

Calculate the matrix elements

$$\langle m | \hat{x} | n \rangle.$$

(b)

Calculate

$$\langle m | \hat{p} | n \rangle.$$

(c)

Construct the matrices representing $\hat{x}$ and $\hat{p}$ using the basis

$$|0\rangle, |1\rangle, |2\rangle, |3\rangle, |4\rangle.$$

(d)

Verify explicitly that

$$[\hat{x}, \hat{p}] = i\hbar I$$

within this truncated basis.

How does the accuracy of the commutator change as the basis size is increased?

(e)

From your results, explain why electric-dipole transitions in the harmonic oscillator satisfy the selection rule

$$\Delta n = \pm 1.$$

SOLUTION:

Problem 3 — The Harmonic Oscillator Wavefunctions

Although the energy spectrum of the harmonic oscillator can be obtained entirely using the ladder-operator formalism, it is useful to examine the corresponding wavefunctions in position space.

The normalized stationary-state wavefunctions are

$$\psi_n(x) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi \hbar} \right)^{1/4} H_n(\xi) e^{-\xi^2/2},$$

where

$$\xi = \sqrt{\frac{m\omega}{\hbar}} x$$

and $H_n(\xi)$ is the Hermite polynomial of degree n .

Unless otherwise stated, use dimensionless harmonic-oscillator units,

$$\hbar = m = \omega = 1.$$

Physical Objective

Construct and visualize the stationary-state wavefunctions of the quantum harmonic oscillator. Examine how their shape, parity, and spatial extent change with increasing energy.

Mathematical Background

The energy eigenvalues are

$$E_n = n + \frac{1}{2}, \quad n = 0, 1, 2, \dots$$

in dimensionless units.

The classical turning points satisfy

$$\frac{1}{2}x_{\text{tp}}^2 = E_n,$$

or

$$x_{\text{tp}} = \pm \sqrt{2E_n}.$$

As the energy increases, the wavefunctions become more oscillatory and extend farther from the origin.

Computational Tasks

(a) Plot the first six stationary states

Plot

$$\psi_n(x), \quad n = 0, 1, \dots, 5,$$

on the same figure.

(b) Plot the corresponding probability densities

Plot

$$|\psi_n(x)|^2$$

for the same six states. What differences do you observe between the wavefunctions and the probability densities?

(c) Energy-level diagram

Construct a figure showing

- the harmonic oscillator potential,

$$V(x) = \frac{1}{2}x^2,$$

- the first six energy levels,

$$E_n = n + \frac{1}{2},$$

- and the corresponding wavefunctions drawn at their respective energies. This is one of the standard visualizations of the harmonic oscillator.

(d) Classical turning points

For each state, compute the classical turning points

$$x_{\text{tp}} = \pm\sqrt{2E_n}.$$

Indicate these turning points on your probability-density plots.

How much of the quantum probability density lies outside the classically allowed region?

(e) Nodes and parity

Determine the number of nodes for each wavefunction.

Verify that

- even values of n produce even wavefunctions,

$$\psi_n(-x) = \psi_n(x),$$

- odd values of n produce odd wavefunctions,

$$\psi_n(-x) = -\psi_n(x).$$

Expected Output

Your notebook should include

- plots of the first six wavefunctions,
- plots of the first six probability densities,
- an energy-level diagram with wavefunctions,
- classical turning points for each state.

Physical Interpretation

Answer the following questions.

1. How does the number of nodes depend on the quantum number?
 2. Why does the spatial extent of the wavefunction increase with increasing energy?
 3. Why is there a finite probability of finding the particle outside the classical turning points?
 4. How does the harmonic oscillator differ from the infinite square well?
 5. Which features of the wavefunctions become increasingly "classical" as n becomes large?
-

★ Challenge

The Hermite polynomials satisfy the recursion relation

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x).$$

Write your own Python routine that generates the Hermite polynomials recursively instead of using `scipy.special.eval_hermite`.

Compare your results with SciPy for the first ten wavefunctions.

SOLUTION:

Problem 4 — Orthogonality and Basis Convergence

The harmonic-oscillator eigenfunctions form an orthonormal set:

$$\int_{-\infty}^{\infty} \psi_m^*(x) \psi_n(x) dx = \delta_{mn}.$$

In numerical calculations, however, the integral must be evaluated over a finite spatial interval using a finite grid. The result therefore depends on both the size of the computational domain and the grid resolution.

Physical Objective

Verify the normalization and orthogonality of the harmonic-oscillator eigenfunctions numerically. Examine how the accuracy of the calculation depends on the spatial range

and grid spacing.

Mathematical Background

Define the overlap matrix

$$S_{mn} = \int_{-\infty}^{\infty} \psi_m^*(x) \psi_n(x) dx.$$

For an exact orthonormal basis,

$$S_{mn} = \delta_{mn}.$$

Thus, the exact overlap matrix is the identity matrix:

$$S = I.$$

In a numerical calculation, the integral is approximated over a finite interval,

$$-x_{\max} \leq x \leq x_{\max},$$

and evaluated on a grid containing N_x points.

The numerical overlap matrix will therefore differ slightly from the identity.

Computational Tasks

(a) Verify normalization

For the first six harmonic-oscillator states, calculate

$$\int_{-x_{\max}}^{x_{\max}} |\psi_n(x)|^2 dx.$$

Use an appropriate numerical integration routine, such as

`scipy.integrate.simpson`.

Report the normalization integral for each state.

(b) Construct the overlap matrix

Calculate

$$S_{mn} = \int_{-x_{\max}}^{x_{\max}} \psi_m^*(x) \psi_n(x) dx$$

for

$$m, n = 0, 1, \dots, 5.$$

Display the result as a numerical matrix.

The matrix should be approximately equal to the identity matrix.

(c) Measure the numerical error

Define the error matrix

$$\Delta S = S - I.$$

Calculate the maximum absolute error,

$$\epsilon_{\max} = \max_{m,n} |S_{mn} - \delta_{mn}|.$$

Also calculate the Frobenius norm,

$$\epsilon_F = \sqrt{\sum_{m,n} |S_{mn} - \delta_{mn}|^2}.$$

Report both errors.

(d) Convergence with spatial range

Hold the number of grid points fixed and repeat the calculation for several values of $x_{\max}$.

For example,

`x_max_values = [3.0, 4.0, 5.0, 6.0, 8.0]`

For each value of $x_{\max}$, calculate $\epsilon_{\max}$.

Plot

$$\epsilon_{\max}$$

as a function of $x_{\max}$.

Use a logarithmic scale for the vertical axis if useful.

(e) Convergence with grid resolution

Hold the spatial range fixed and repeat the calculation for several values of N_x .

For example,

$N_x_values = [101, 201, 501, 1001, 2001]$

For each value of N_x , calculate $\epsilon_{\max}$.

Plot the numerical error as a function of N_x .

Expected Output

Your notebook should include

- a table of normalization integrals,
 - the numerical overlap matrix,
 - the overlap-error matrix,
 - the maximum and Frobenius errors,
 - a convergence plot with respect to $x_{\max}$,
 - a convergence plot with respect to N_x .
-

Physical Interpretation

Answer the following questions.

1. Why does a spatial interval that is too small produce inaccurate normalization integrals?
 2. Which states are most sensitive to the choice of $x_{\max}$? Explain.
 3. Why do higher-energy states generally require a larger spatial domain?
 4. Does increasing the number of grid points always improve the result?
 5. What limits the accuracy once the grid is sufficiently fine?
 6. How can you determine that a numerical result has converged?
-

Basis Completeness

Orthogonality and completeness are related but distinct properties.

Orthogonality means that

$$\langle m | n \rangle = \delta_{mn}.$$

Completeness means that an arbitrary physical state can be represented as

$$|\Psi\rangle = \sum_{n=0}^{\infty} c_n |n\rangle.$$

A finite numerical basis contains only the first N eigenstates:

$$|\Psi\rangle \approx \sum_{n=0}^{N-1} c_n |n\rangle.$$

The accuracy of this approximation must be tested by increasing N until the quantities of interest no longer change significantly.

★ Challenge — Resolution of the Identity

Within a truncated basis, define

$$K_N(x, x') = \sum_{n=0}^{N-1} \psi_n(x) \psi_n^*(x').$$

In a complete basis,

$$\sum_{n=0}^{\infty} \psi_n(x) \psi_n^*(x') = \delta(x - x').$$

Choose a fixed value $x' = x_0$ and plot

$$K_N(x, x_0)$$

for several values of N .

Describe how the function changes as the basis size increases.

Explain why the finite-basis result cannot be an exact Dirac delta function.

SOLUTION:

Problem 5 — Expectation Values and the Uncertainty Principle

The ladder-operator formalism allows many expectation values to be evaluated without directly performing integrals. In this problem, you will use the operator expressions for

position and momentum to calculate expectation values and uncertainties for the harmonic-oscillator stationary states.

Physical Objective

Determine the position and momentum expectation values for the harmonic-oscillator eigenstates. Use these results to examine the Heisenberg uncertainty relation and the role of zero-point motion.

Mathematical Background

The position and momentum operators may be written as

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger),$$

and

$$\hat{p} = -i\sqrt{\frac{m\hbar\omega}{2}} (\hat{a} - \hat{a}^\dagger).$$

The ladder operators act on the energy eigenstates according to

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle,$$

and

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle.$$

The uncertainty in an observable A is

$$\Delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}.$$

The Heisenberg uncertainty relation is

$$\Delta x \Delta p \geq \frac{\hbar}{2}.$$

Analytical Tasks

(a) Position expectation value

Use the ladder-operator representation of $\hat{x}$ to show that

$$\langle n | \hat{x} | n \rangle = 0.$$

Explain why this result is also expected from the parity of the stationary states.

(b) Momentum expectation value

Show that

$$\langle n | \hat{p} | n \rangle = 0.$$

Give both an operator-based and a physical explanation.

(c) Position variance

Expand $\hat{x}^2$ in terms of the ladder operators and show that

$$\langle n | \hat{x}^2 | n \rangle = \frac{\hbar}{m\omega} \left(n + \frac{1}{2} \right).$$

Hence show that

$$\Delta x = \sqrt{\frac{\hbar}{m\omega} \left(n + \frac{1}{2} \right)}.$$

(d) Momentum variance

Show that

$$\langle n | \hat{p}^2 | n \rangle = m\hbar\omega \left(n + \frac{1}{2} \right).$$

Hence obtain

$$\Delta p = \sqrt{m\hbar\omega \left(n + \frac{1}{2} \right)}.$$

(e) Uncertainty product

Show that

$$\Delta x \Delta p = \hbar \left(n + \frac{1}{2} \right).$$

Which stationary state saturates the Heisenberg uncertainty bound?

Computational Tasks

Use dimensionless harmonic-oscillator units,

$$\hbar = m = \omega = 1.$$

For the first six stationary states, calculate numerically

$$\langle x \rangle, \quad \langle x^2 \rangle, \quad \langle p \rangle, \quad \langle p^2 \rangle,$$

and then determine

$$\Delta x, \quad \Delta p, \quad \Delta x \Delta p.$$

You may evaluate the expectation values directly in position space.

For example,

$$\langle x \rangle = \int_{-\infty}^{\infty} \psi_n^*(x) x \psi_n(x) dx,$$

and

$$\langle p \rangle = \int_{-\infty}^{\infty} \psi_n^*(x) \left(-i \frac{d}{dx} \right) \psi_n(x) dx.$$

For the second moments,

$$\langle p^2 \rangle = \int_{-\infty}^{\infty} \psi_n^*(x) \left(-\frac{d^2}{dx^2} \right) \psi_n(x) dx.$$

Use numerical differentiation and integration carefully.

Expected Output

Construct a table containing

n	$\langle x \rangle$	$\langle x^2 \rangle$	$\langle p \rangle$	$\langle p^2 \rangle$	Δx	Δp	$\Delta x \Delta p$
0							
1							
2							
3							
4							
5							

Compare your numerical values with the analytical results.

Energy Decomposition

The kinetic and potential energies are

$$\langle T \rangle = \frac{\langle p^2 \rangle}{2m},$$

and

$$\langle V \rangle = \frac{1}{2}m\omega^2 \langle x^2 \rangle.$$

(f)

Show that, for every stationary state,

$$\langle T \rangle = \langle V \rangle = \frac{1}{2} E_n.$$

This is a special case of the virial theorem.

Verify this result numerically for the first six states.

Physical Interpretation

Answer the following questions.

1. Why are $\langle x \rangle$ and $\langle p \rangle$ both zero for every stationary state?
2. Does $\langle x \rangle = 0$ imply that the particle is located at the origin?
3. Why do Δx and Δp increase with n ?
4. What does the nonzero value of Δx in the ground state tell you about zero-point motion?
5. Why does the ground state minimize the uncertainty product?
6. How are the uncertainty product and the oscillator energy related?

★ Challenge — Matrix Calculation

Construct finite-dimensional matrix representations of

$$\hat{a}, \quad \hat{a}^\dagger, \quad \hat{x}, \quad \hat{p},$$

using the first N harmonic-oscillator states.

Use matrix multiplication to calculate

$$\langle n | \hat{x}^2 | n \rangle, \quad \langle n | \hat{p}^2 | n \rangle,$$

and compare with the analytical expressions.

Investigate how truncating the basis affects the results for states near the upper edge of the basis.

SOLUTION:

Problem 6 — The Variational Method

The harmonic oscillator is one of the few quantum systems that can be solved exactly. Most quantum systems, however, do not possess exact analytical solutions. One of the most powerful approximation methods in quantum mechanics is the **variational method**, which provides an upper bound to the ground-state energy.

In this problem, you will use a Gaussian trial wavefunction to estimate the ground-state energy of the harmonic oscillator.

Physical Objective

Use a trial wavefunction containing an adjustable parameter to estimate the ground-state energy of the harmonic oscillator. Determine the optimal wavefunction by minimizing the expectation value of the Hamiltonian.

Mathematical Background

The variational theorem states that

$$E[\phi] = \frac{\langle \phi | \hat{H} | \phi \rangle}{\langle \phi | \phi \rangle} \geq E_0,$$

where E_0 is the exact ground-state energy.

Consider the normalized Gaussian trial function

$$\phi_\alpha(x) = \left(\frac{2\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2}, \quad \alpha > 0.$$

The parameter α controls the width of the Gaussian.

(a) Analytical Calculation

Show that

$$\langle T \rangle = \frac{\hbar^2 \alpha}{2m},$$

and

$$\langle V \rangle = \frac{m\omega^2}{8\alpha}.$$

Hence show that

$$E(\alpha) = \frac{\hbar^2 \alpha}{2m} + \frac{m\omega^2}{8\alpha}.$$

(b) Find the Optimum Width

Differentiate $E(\alpha)$ with respect to α and determine the value of α that minimizes the energy.

Calculate the corresponding minimum energy.

Compare your result with the exact ground-state energy.

(c) Numerical Investigation

Write a Python function that evaluates

`E(alpha)`

for any positive value of `alpha`.

Plot

- the kinetic energy,
- the potential energy,
- the total energy,

as functions of α .

Indicate the minimum on the graph.

Use a numerical optimization routine (such as `scipy.optimize.minimize_scalar`) to locate the minimum and compare it with your analytical result.

Physical Interpretation

Answer the following questions.

1. Why does the kinetic energy increase as the Gaussian becomes narrower?
 2. Why does the potential energy increase as the Gaussian becomes broader?
 3. Why must the total energy possess a minimum?
 4. Why does the variational method give the exact answer in this case?
 5. Explain why the variational theorem is useful even for systems that cannot be solved exactly.
-

★ Challenge

Repeat the calculation for the anharmonic potential

$$V(x) = \frac{1}{2}x^2 + \lambda x^4.$$

Using the same Gaussian trial function, evaluate the energy numerically for several values of $\lambda > 0$.

Does the Gaussian remain an exact solution?

How does the optimal width change as λ increases?

SOLUTION:
